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Fine-grinding characteristics of hard materials by attrition mill

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Abstract

The fine grinding characteristics of hard materials such as synthetic diamond and alumina were investigated by means of an attrition mill. The grinding kinetics approach was successfully applied to the analysis of product size distributions obtained under various process conditions. As a result, the higher grinding rate of hard materials was proven to be obtained with the larger media size, the higher impeller rotational velocity and the larger amount of the feed within the studied range. It was also found that the volume of the dispersing liquid has an optimal value for the grinding of alumina. © 1999 Elsevier Science S.A. All rights reserved.

Keywords: Comminution; Attrition mill; Hard fine materials; Grinding kinetics

1. Introduction

Stirred-ball attrition mills are widely used for fine grinding of various materials. They are supposed to be most efficient among grinding media-based compression-shear type mills. The main advantages of the attrition mills are the relatively high energy utilization, fast and efficient fine grinding, proficient temperature control and simple operation [1]. The feed material is comminuted among the moving media, the stirrer, and the grinding chamber wall by friction, impact, and compression forces.

The objective of the present study is to investigate the fine grinding characteristics of hard materials by means of the attrition mill. The grinding-kinetics approach is adopted to analyze the particle size distribution of products obtained under various process conditions.

2. Experiments

The grinding tests were performed in a vertical stirred-ball mill MA01D manufactured by Mitsui Mining (Japan). Fig. 1 illustrates the schematics of a laboratory-scale batch type of the attrition mill. The mill consists of a stationary stainless steel chamber holding a centrally positioned rotating stirrer system with a five-arm impeller fitted to the stirrer shaft. The cylindrical pins of 6 mm diameter and 65 mm length are arranged at 90° to each other. The net volume of the milling chamber is 0.8 l. The chamber is filled with spherical steel balls of average diameters 2.00, 4.76 and 6.36 mm which are accelerated by the motion of the stirrer. The grinding media occupies 80% of the chamber volume. The mill is equipped with a frequency transformer, and the speed of the electrical drive can be set in the range between 90 and 528 rpm. A water bath is attached for temperature control.

The hard materials such as alumina CBA10 obtained from Showa Denko (Japan) and synthetic diamond supplied by Izumi Tech. (Japan) were used for grinding experiments. The quartz sand KCM from Kyoritsu Ceramic Materials (Japan) was also used for comparison as a relatively soft material. The water was adopted as liquid dispersing media.

The mill was run for 8 h with regular interruptions to take samples using a syringe. The samples collected during milling were treated with condense hydrochloric acid in order to remove contaminants derived from the mill chamber, the impeller arms and the grinding media. The particle size distribution was measured by a laser diffraction and scattering method (SALD 2000, Shimadzu).

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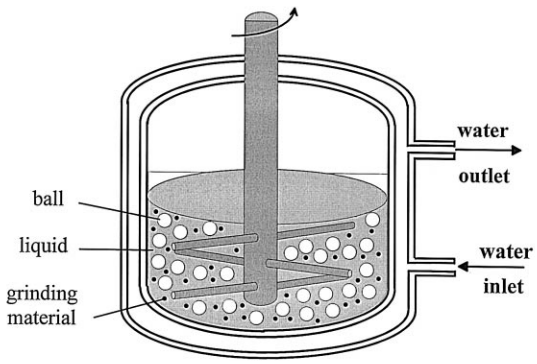

Fig. 1. Schematic drawing of laboratory stirred-ball mill.

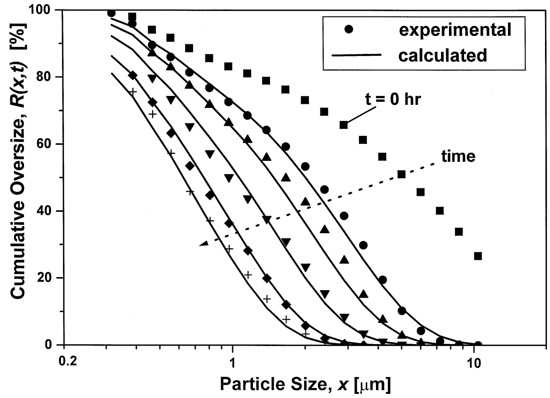

Fig. 2. Diamond product size distributions for different grinding time ($d_b = 4.76$ mm, $V_r = 412$ rpm, $V_l = 100$ cm^3, $W = 50$ g).

Four sets of experiments were conducted to determine the effects of the ball size, the impeller speed, the liquid volume and the charge weight on the size distribution of the product at different grinding time.

3. Modeling of grinding kinetics

The kinetics of comminution is modeled by using a classical batch grinding equation based on selection and breakage functions. The change in particle-size distribution of ground material is to be described as a function of grinding time, t, as follows [2]:

$$\frac{\partial^2 D(x,t)}{\partial t \partial x} = -S(x)\frac{\partial D(x,t)}{\partial x} + \int_x^{x_m} \frac{\partial D(\gamma,t)}{\partial \gamma} S(\gamma) \frac{\partial B(\gamma,x)}{\partial x} \mathrm{d}\gamma \qquad (1)$$

where $D(x,t)$ is the cumulative undersize of particles, x is the particle size, γ is the size of a single particle to be broken, and x_m is the maximum particle size present.

$S(x)$ is the selection function which represents the probability for a particle of size x to be selected for grinding, and is specified by the empirical relationship as:

$$S(x) = Kx^n, \qquad (2)$$

where K is a grinding rate constant and n is a constant. $B(\gamma,x)$ is the breakage function which describes the distribution of fragment sizes produced from the initial particle of size γ, and is written as:

$$B(\gamma,x) = \left(\frac{x}{\gamma}\right)^m, \qquad (3)$$

where m is a constant.

The following analytical solution of the batch grinding Eq. (1) was obtained from substitution of Eqs. (2) and (3) by Nakajima and Tanaka [3]:

$$R(x,t) \approx R(x,0)\exp\left[-(\mu K x^n t)^\nu\right], \qquad (4)$$

$$R(x,t) = R(x,0)\exp\left[-Kx^n t\right] \qquad \text{for } m = n \qquad (5)$$

where $R(x,0)$ is the initial oversize distribution function. Constants μ and ν are determined only by the value of ratio m/n on a transformation chart [3]. Thus, the rate constant K and exponents m and n can be estimated by fitting experimental data with Eq. (4).

For this purpose, Eq. (4) is rewritten in the following linear form with respect to ln(t) and ln(x):

$$\ln\left[-\ln\left(\frac{R(x,t)}{R(x,0)}\right)\right] = \nu\ln(\mu K t) + n\nu\ln(x) \qquad (6)$$

where

$$\nu\ln(\mu K t) = \nu\ln(\mu K) + \nu\ln(t) \qquad (7)$$

Thus, linear least-square approximation of experimental data by Eq. (6) yields the values of $\nu\ln(\mu Kt)$ and $n\nu$. Then, ν is given by the plot of Eq. (7) and μ is read through the ratio m/n and μ^ν for ν on the chart. Finally, the grinding rate constant K can be calculated from the value of the intercept for corresponding ν and μ^ν.

4. Results and discussions

Fig. 2 illustrates the typical size distributions for the diamond obtained at various grinding times in the attrition mill. The corresponding dependence of the median size on the grinding time is shown in Fig. 3. The median size of

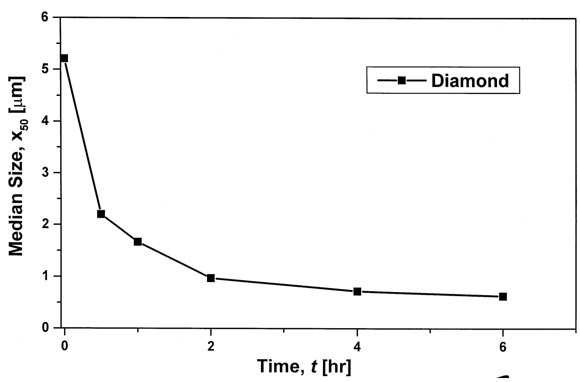

Fig. 3. Variation of product median size with grinding time.

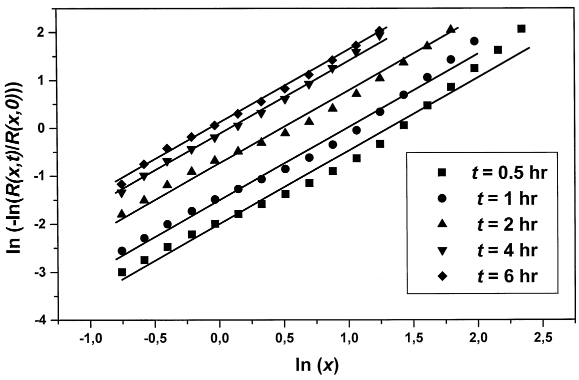

Fig. 4. Relationship between normalized oversize and particle size.

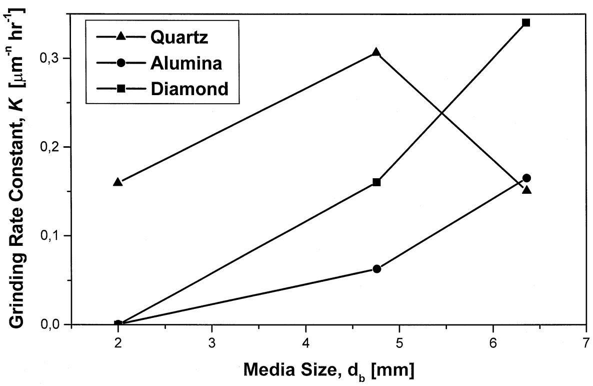

Fig. 6. Effect of media size on grinding rate constant ($V_l = 200$ cm^3, $V_r = 412$ rpm, $W = 100$ g).

the diamond decreases with increasing time and gets smaller than 1 μm in 2 h. Then, a regression analysis was made on the product size distributions. Fig. 4 shows the linear dependence of normalized cumulative oversize functions on particle size in logarithmic scale for different grinding times. The intersects of straight lines on the ordinate are plotted against the grinding time in Fig. 5. The parameters of grinding kinetics Eq. (4) obtained by the linear regression analysis are used to reproduce the product size distributions in Fig. 2. The good agreement between experimental and calculated size distributions confirms the applicability of the rate Eq. (4) to the description of fine grinding of hard materials.

The effect of the media size on the product size distribution of different materials was studied by conducting a series of experiments with differently sized steel balls, while other conditions were maintained constant. Grinding rate constants for alumina, diamond and quartz sand are shown in Fig. 6 as a function of the media size. They increased with the media size for alumina and diamond particles. This fact is contradictory to the findings reported on the grinding of the ordinary materials [5].

To understand the reason for the opposite dependency of the rate constants on the media size, the dominant mechanism of comminution in the attrition mill should be investigated. The abrasion, compression and impact fracture are identified as main breakage mechanisms [6]. The particle size distribution of a material after comminution will be determined by the combination of all mechanisms. The comparative contribution of the particular mechanism will depend on the operational conditions and the material properties being ground. The particle attrition is often assumed as a primary rupture mechanism in a stirred-ball attrition mill. However, Varinot et al. [7] pointed out on the relative importance of the impact fracture to abrasion in fine grinding with the attrition mill. In case of the communition of the hard fine materials, the impact mechanism can be expected to play the dominant role in breakage, because the hard materials will be difficult to be abraded on their surfaces. Fig. 7 illustrates the particle size distributions of the product collected after 6 h of grinding. The bimodal distribution curve corresponds to the abrasion mechanism, and the impact fracture results in the relatively wide size distribution. As this figure shows, the relative contribution of impact in comparison with abrasion increases in the following order: quartz–alumina–diamond. Then, it can be expected that increasing the media size the contribution of the impact mechanism will increase for hard materials. Thus, the grinding rate constant is greater

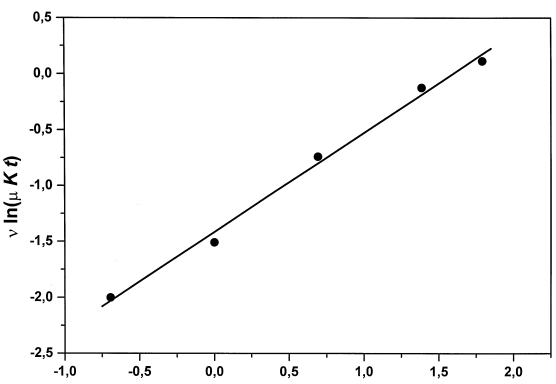

Fig. 5. Relationship between intercept value in Fig. 4 and grinding time.

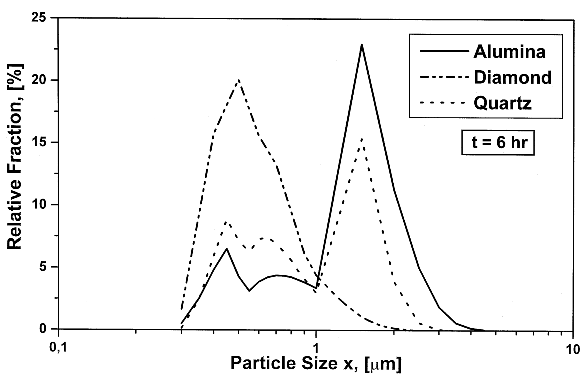

Fig. 7. Product size distribution for different feed materials ($t = 6$ h, $d_b = 4.76$ mm, $V_l = 200$ cm^3, $V_r = 412$ rpm, $W = 100$ g).

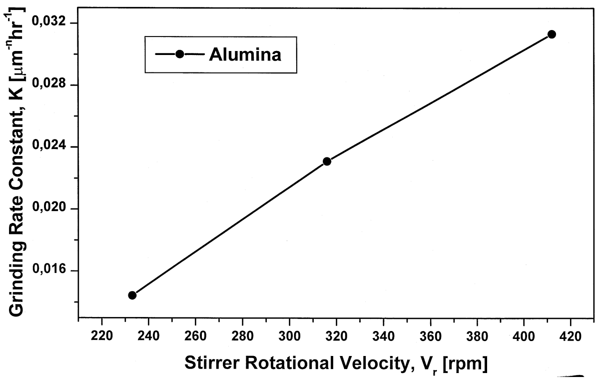

Fig. 8. Effect of stirrer rotational velocity on grinding rate constant ($d_b = 4.76$ mm, $V_l = 200$ cm^3, $W = 50$ g).

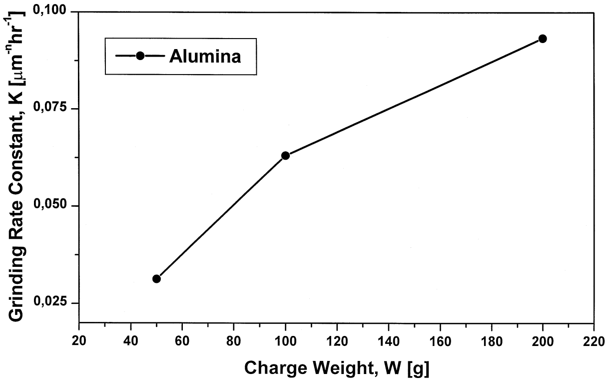

Fig. 10. Effect of charge weight on grinding rate constant ($d_b = 4.76$ mm, $V_l = 200$ cm^3, $V_r = 412$ rpm).

for the larger media, because larger balls are supposed to hit particles with higher kinetic energy and break them faster. Concerning the grinding of ordinary materials like quartz sand, when the abrasion is the prevailing mechanism of comminution with the attrition mill, the grinding rate gets higher with decreasing media size due to larger number of media balls available for grinding. This trend continues until the media size becomes too small to cause the particle abrasion effectively. And then, K decreases with smaller media size due to smaller impact effect. The results for quartz sand agree with those reported by Zheng et al. [4] for grinding limestone.

The effect of impeller rotational speed on grinding of alumina is shown in Fig. 8. The grinding rate constant increases with the rotational speed. The kinetic energy and the number of collisions of the media balls will be higher with the rapidly rotated impeller and thus the desired product distribution could be obtained in shorter grinding time.

Fig. 9 illustrates the effect of the liquid volume on the grinding rate constant for the alumina and the diamond. The media size, the impeller speed and the charge weight remained the same during the experiments. The rate constant for the alumina was found to increase with the addition of liquid from 0 to 200 cm^3 and then decrease at 400 cm^3. As the amount of water increases, the viscosity of the slurry decreases and the slurry behaves sufficiently like fluid, which results in grinding improvements due to smaller resistance to the balls and higher effective dispersion volume of the particles inside the pot. Nevertheless, further dilution of the suspension may lead to fewer and weaker interactions between balls and particles within the liquid [7]. Thus, the reduction of probability for the particle to be selected for grinding reflects the decrease in the rate constant. The grinding rate constant for the diamond is lower in the liquid volume of 200 cm^3 than that in 100 cm^3. The results for dry grinding for alumina are also shown in this figure. In comparison with wet grinding, dry grinding results in broader product distribution and lower production rate of fines due to poor dispersion of particles.

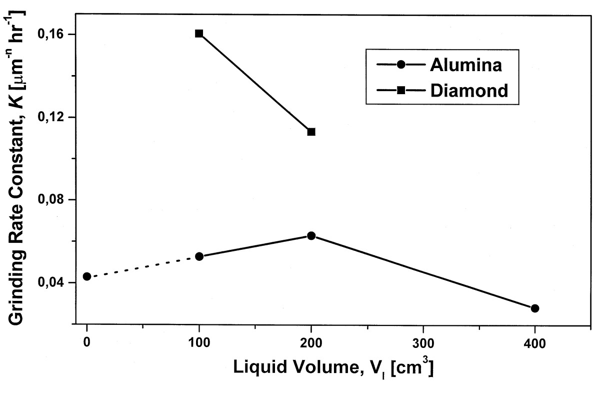

Fig. 9. Effect of liquid volume on grinding rate constant ($d_b = 4.76$ mm, $V_r = 412$ rpm, $W = 100$ g).

The effect of alumina charge weight on the grinding rate constant is shown in Fig. 10 for the same volume of the liquid. The results illustrate that the rate constant increases with the charge weight over the studied range. Apparently, the probability of the particles to be selected for grinding increases with the charge weight assuming the uniform distribution of particles inside the pot. Nevertheless, it can be expected that the rate constant will decrease with subsequent increase in charge due to increase in the slurry viscosity and dissipated impact energy of the balls on each particle. Zheng et al. [8] also speculated that at high solids concentration the slurry flow pattern inside the mill is not favorable for grinding. That is, the solids are expelled from the active volume between the pin tips and the tank walls and form a layer at the wall or above the pins.

5. Conclusions

It was actually found possible to produce sub-micron hard particles by means of the stirred-ball attrition mill. The dominant breakage mechanism should be based on the impact effect. The grinding kinetics approach was successfully applied to the analysis of product size distribution. As the result of the experimental parametric analysis, the grinding rate constants for the alumina and diamond were found to increase with the media size, the impeller rotational speed and the feed charge weight within the studied range. But, the dependence of the rate constant on liquid volume and charge weight for the alumina seems to go through a maximum.

6. Nomenclature

$B(\gamma,x)$	breakage function of particles $(-)$
d_b	media ball diameter (mm)
$D(x,t)$	cumulative undersize of particles $(-)$
K	grinding rate constant in selection function (μm^{-n} h^{-1})
m	constant in breakage function $(-)$
n	constant in selection function $(-)$
$R(x,t)$	cumulative oversize of particles $(-)$
$S(x)$	selection function of particles (h^{-1})
t	grinding time (h)
V_l	liquid volume in grinding pot (cm^3)
V_r	impeller rotational speed (rpm)
W	charge weight of ground material (g)
x	particle size (μm)
x_m	maximum particle size (μm)
γ	particle size to be broken (μm)
μ, ν	parameters determined from ratio m/n on a chart $(-)$

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